

Smart Food Delivery System with Priority Dispatch

- PRD

1. Product Overview

A real-time food delivery system that assigns delivery partners efficiently using advanced data structures. It supports priority orders, batching, and dynamic reassignment.

2. Objectives

- 1 Assign nearest delivery partner
- 2 Handle priority orders efficiently
- 3 Enable order batching
- 4 Support dynamic reassignment
- 5 Demonstrate advanced data structures

3. Users

- 1 Customers
- 2 Delivery partners
- 3 Admin (simulation monitoring)

4. Core Features

4.1 Nearest Partner Assignment

Uses KD-Tree for efficient nearest neighbor search.

4.2 Priority Handling

Uses a heap-based priority queue to process priority orders first.

4.3 Order Batching

Groups nearby orders using spatial queries.

4.4 Dynamic Reassignment

Reassigns delivery partners when a higher priority order arrives.

5. Data Structures

- 1 KD-Tree for spatial search
- 2 Heap for priority handling
- 3 Balanced BST for partner management

6. Workflow

- 1 Order placed
- 2 Classified into priority or normal
- 3 Assigned to nearest partner
- 4 Batching applied
- 5 Reassignment triggered if needed

7. Functional Requirements

- 1 Add delivery partners
- 2 Place orders
- 3 Auto-assign partners
- 4 Batch orders
- 5 Reassign dynamically

8. Non-Functional Requirements

- 1 High performance
- 2 Scalable system
- 3 Reliable assignments
- 4 User-friendly interface

9. Data Models

- 1 Order: id, location, priority, status
- 2 Partner: id, location, status
- 3 Batch: list of orders

10. Success Metrics

- 1 Reduced delivery time
- 2 Faster priority deliveries
- 3 Efficient batching rate

11. Edge Cases

- 1 No available partners
- 2 Multiple priority orders
- 3 Reassignment conflicts

12. Future Enhancements

- 1 Route optimization
- 2 Real-time tracking
- 3 ML-based predictions